

QMST5336: ANALYTICS

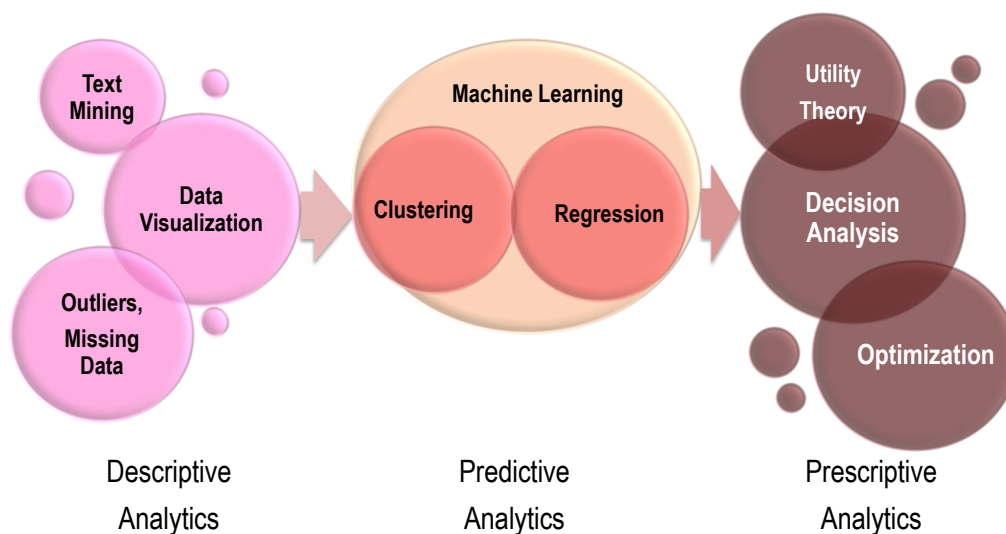
Lecture 6: Regression I: Linear Regression

Emily Zhu, PhD.

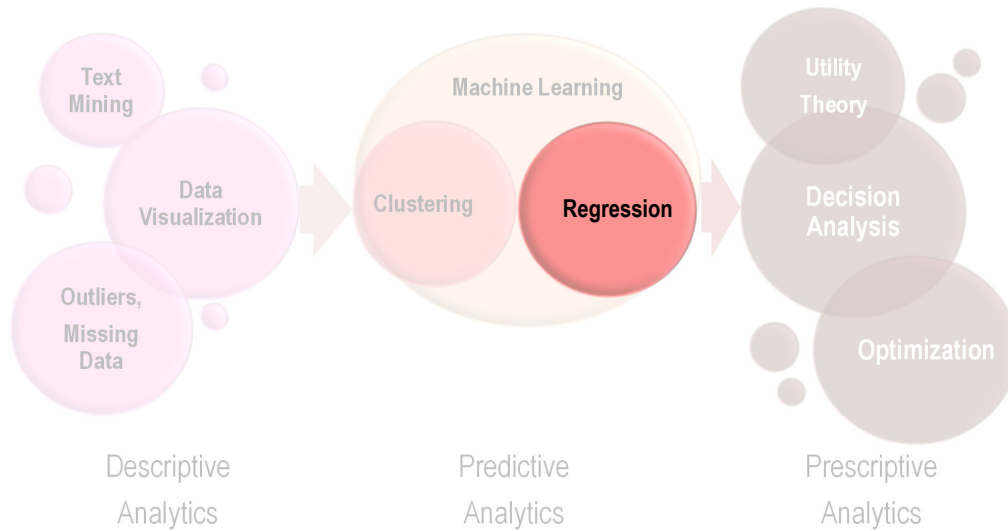
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Our Problem and Roadmap

An investment property in Austin: **Airbnb** versus **Rental**?

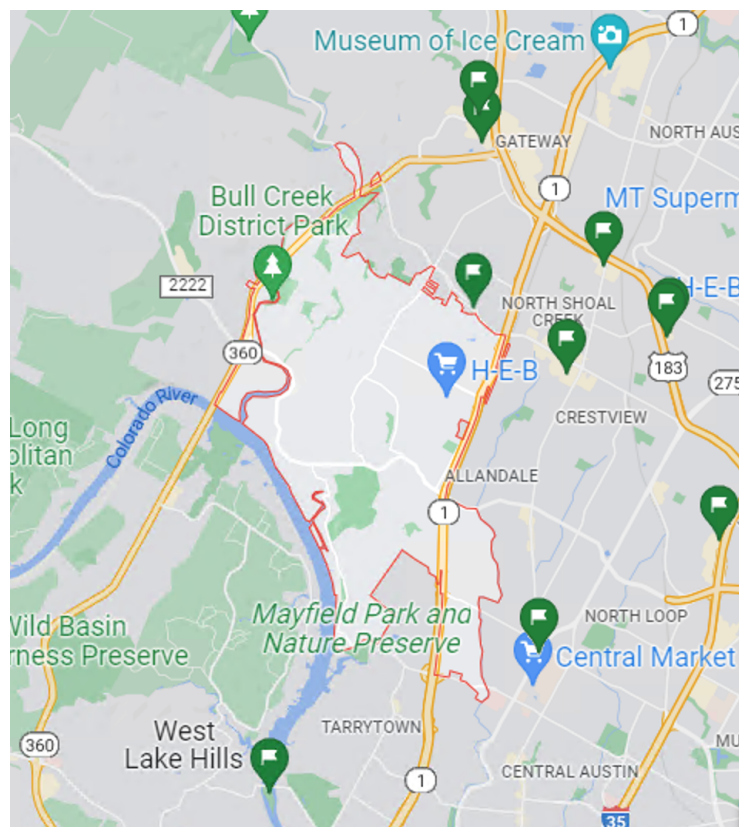


Roadmap: Where We Are



Case: How Much to List the House in North Austin

Suppose the investment property is a house located in 78731, north of Austin.

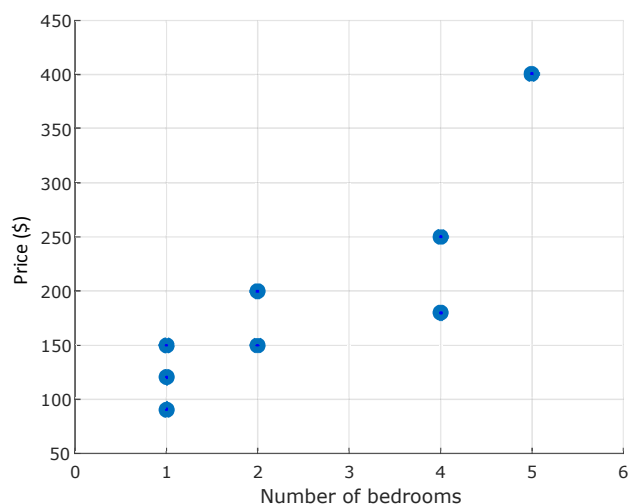


Simple Regression

We would like to understand how:

- x (the independent/explanatory variable) affects y (the dependent variable)
- Example: How does number of bedrooms affects listing price y

Listing	Price y	Bedrooms x
A	\$150	1
B	\$250	4
C	\$150	2
D	\$400	5
E	\$200	2
F	\$120	1
G	\$180	4
H	\$90	1



Linear regression

To predict relationships better, we use multiple regression

- Multiple regression allows us to include multiple explanatory variables in one equation
- This generates better predictions

Simple regression:

$$y = b_0 + b_1x + e$$

Multiple regression:

$$y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_px_p + e$$

We now have p explanatory variables (independent variables)

Find the Best Line in Python

```
1 ### Method 1: Statsmodels ###
2 import statsmodels.formula.api as smf
3 # create a fitted model
4 lm1 = smf.ols(formula = 'Price ~ Bedrooms + Bathrooms', data =
    df_simple).fit()
5 lm1.params
```

```
Intercept 63.333333
Bedrooms 51.666667
Bathrooms 0.2299988
dtype: float64
```

```
1 ###Method 2: Scikit-learn ###
2 from sklearn.linear_model import LinearRegression
3 # create X and Y
4 X_cols = ['Bedrooms', 'Bathrooms']
5 X = df_simple[X_cols]
6 y = df_simple.Price
7 # model fit
8 lm2 = LinearRegression().fit(X, y)
9 # print the coefficients
10 print(lm2.intercept_)
11 print(lm2.coef_)
```

```
63.33333333333333
```

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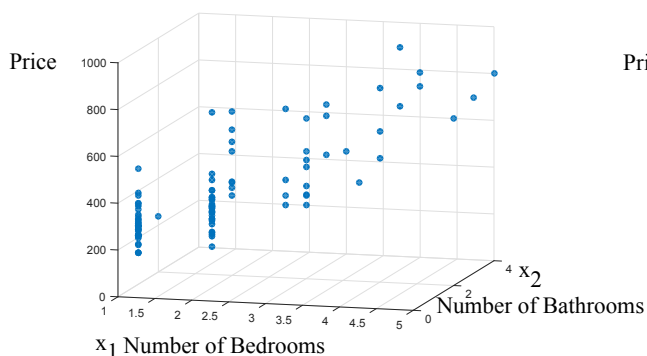
```
array([51.66666667 0.22999988])
```

Visualize Linear Regression

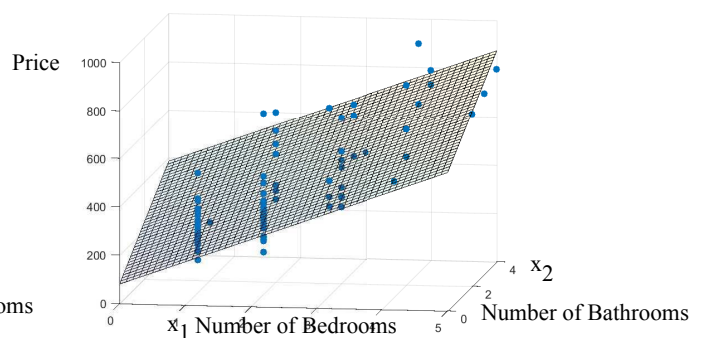
Multiple regression:

$$y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_px_p + e$$

Data



Multiple Regression



Question: How do we interpret this line?

Answer: Each coefficient $\{b_1, b_2, \dots, b_p\}$ represents the effect of a unit increase in $\{x_1, x_2, \dots, x_p\}$, holding everything else equal

Interpret Coefficients of Linear regression

Example:

$$\text{price} = b_0 + b_1(\text{Number of bedrooms}) + b_2(\text{Accommodate}) + b_3(\text{Cleaning fee}) + e$$

- b_1 the effect (on price) caused by adding an extra bedroom to a listing, holding all other factors equal
- b_2 the effect (on price) of accommodate one more person, holding all other factors equal
- b_3 the effect (on price) of one more \$ cleaning fee, holding all other factors equal
- b_0 is the intercept. It represents the price of a house with 0 bedrooms, accommodate 0 person, and 0 charge on cleaning. It might be meaningless to evaluate the regression line at (0,0,0), but we still need it in a regression line.

Prediction: Multiple regression

Example:

$$\begin{aligned}\text{price} = & 98.72 + 84.45(\text{Number of bedrooms}) \\ & + 10.79(\text{Accommodate}) - 0.097(\text{Cleaning fee.}) + e\end{aligned}$$

Suppose I want to list on AirBnb a house :

- with 2 bedrooms
- accommodate up to 5 persons
- a cleaning charge of \$100

How much is the fair listing price on average per night?

$$\begin{aligned}\text{price} &= 98.72 + 84.45(2) + 10.79(5) - 0.097(100) \\ &= 320.60\$\end{aligned}$$

Output Summary

```
1 ### Statsmodels ###
2 lm1.summary()
```

OLS Regression Results

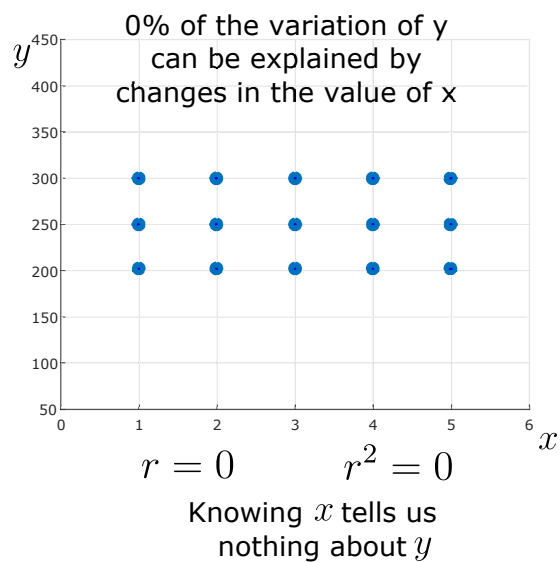
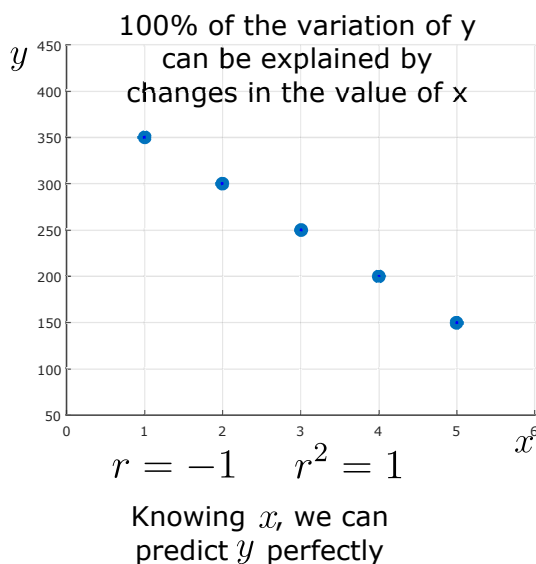
Dep. Variable:	Price	R-squared:	0.729			
Model:	OLS	Adj. R-squared:	0.683			
Method:	Least Squares	F-statistic:	16.11			
Date:	Tue, 08 Oct 2019	Prob (F-statistic):	0.00701			
Time:	10:47:13	Log-Likelihood:	-42.204			
No. Observations:	8	AIC:	88.41			
Df Residuals:	6	BIC:	88.57			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	63.3333	37.534	1.687	0.143	-28.509	155.176
Bedrooms	51.6667	12.874	4.013	0.007	20.165	83.168
Omnibus:	0.542	Durbin-Watson:	1.572			
Prob(Omnibus):	0.763	Jarque-Bera (JB):	0.110			
Skew:	-0.232	Prob(JB):	0.947			
Kurtosis:	2.662	Cond. No.	6.17			

Warnings:

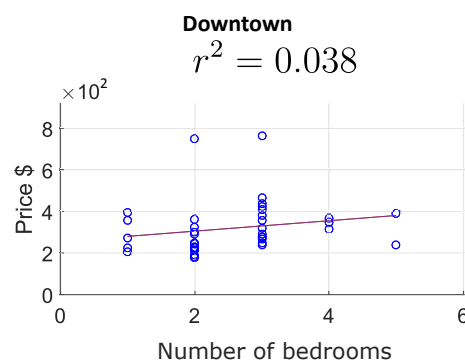
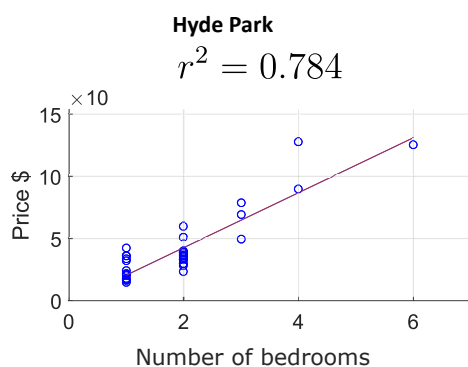
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Statistically, how good is the *Regression Model*?

- The Correlation Coefficient $-1 \leq r \leq 1$
- In regression models $0 \leq r^2 \leq 1$
- The closer r^2 is to 1, the better the line is at showing a relationship



Statistically, how good is the *Regression Model*?



- 78.4% of the variation in listing prices in Hyde Park can be explained by changes in the number of bedrooms
- 3.8% of the variation in listing prices in downtown Austin can be explained by changes in the number of bedrooms

Adjusted R square

The Adjusted R square is a modified version of R square that has been adjusted for the number of explanatory variables in the model.

R square:

- typically go up if we add more explanatory variables

Adjusted R square:

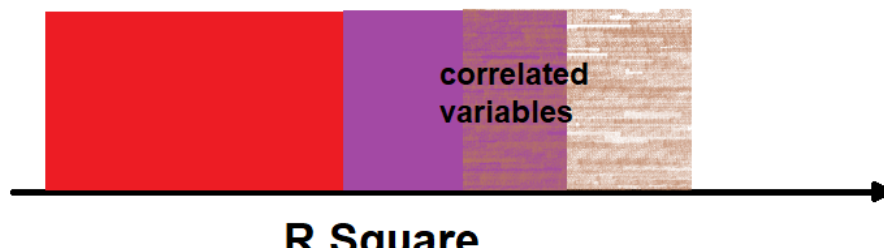
- increases only if the new term improves the model
- It increases when explanatory variable improves model by less than expected by chance
- The adjusted R-squared can be negative, but it's usually not
- It is always lower than the R^2

Adjusted R square help us see when an explanatory variable should be included or excluded from a regression

Adjusted R square: Example

	coef	std err	t	P> t	[0.025	0.975]
Intercept	OLS Regression Results					
accommodates	Dep. Variable:		price		R-squared:	0.526
bathrooms	Model:		OLS		Adj. R-squared:	0.493
bedrooms	Method:		Least Squares		F-statistic:	15.56
Intercept	OLS Regression Results					
accommodates	Dep. Variable:		price		R-squared:	0.522
bedrooms	Model:		OLS		Adj. R-squared:	0.495
	Method:		Least Squares		F-statistic:	19.36

Marginal benefits



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Adjusted R square: Application

Strategy: Unlike R^2 , the adjusted R^2 increases only when the increase in R^2 (due to the inclusion of a new explanatory variable) is more than one would expect to see by chance

- Assume you have a set of explanatory variables (independent variables)
- You order with a hierarchy of importance (e.g. correlation coefficient), and introduce them into a regression one at a time
- Compute the adjusted R^2 computed each time
- the level at which adjusted R^2 reaches a maximum, and decreases afterwards, would be the regression with the ideal combination of having the best fit without excess/unnecessary terms

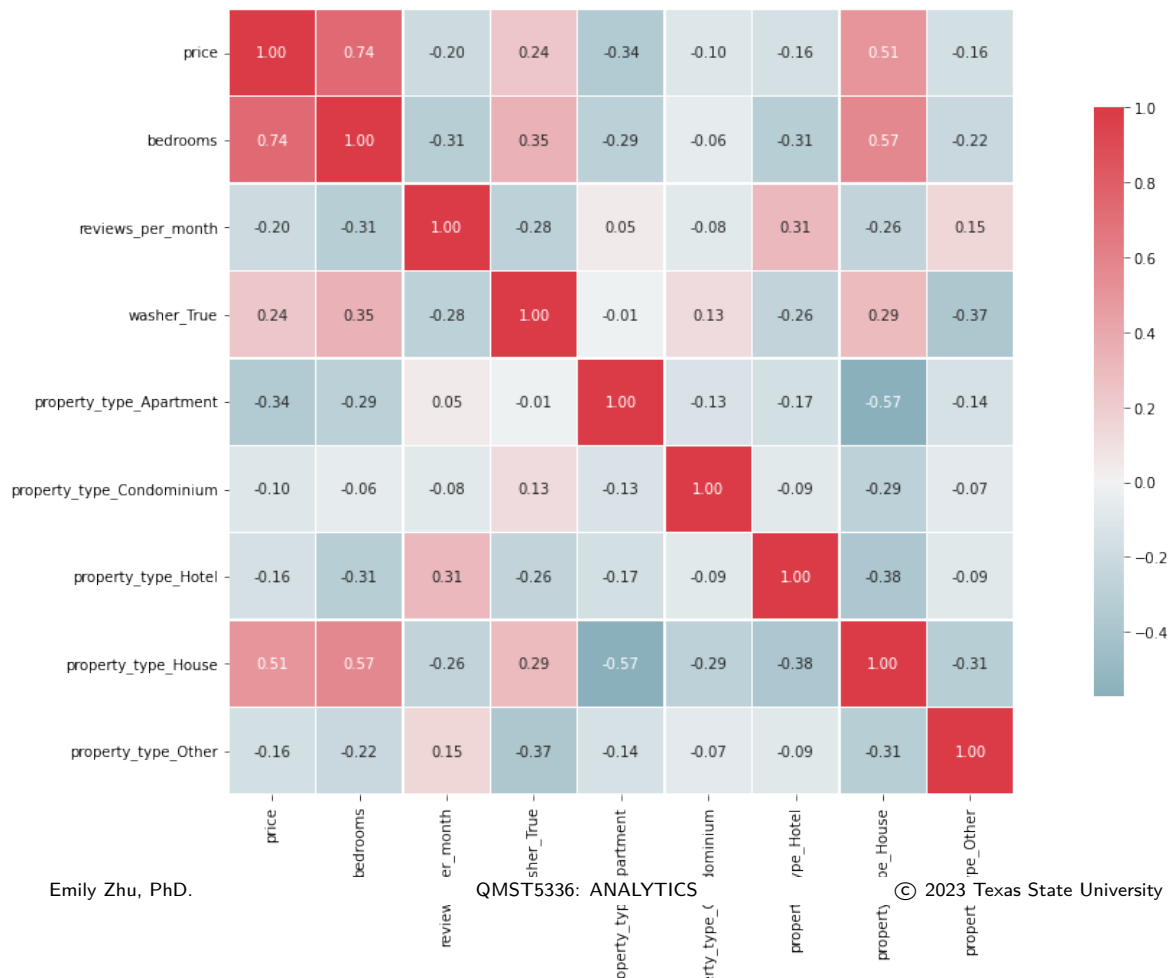
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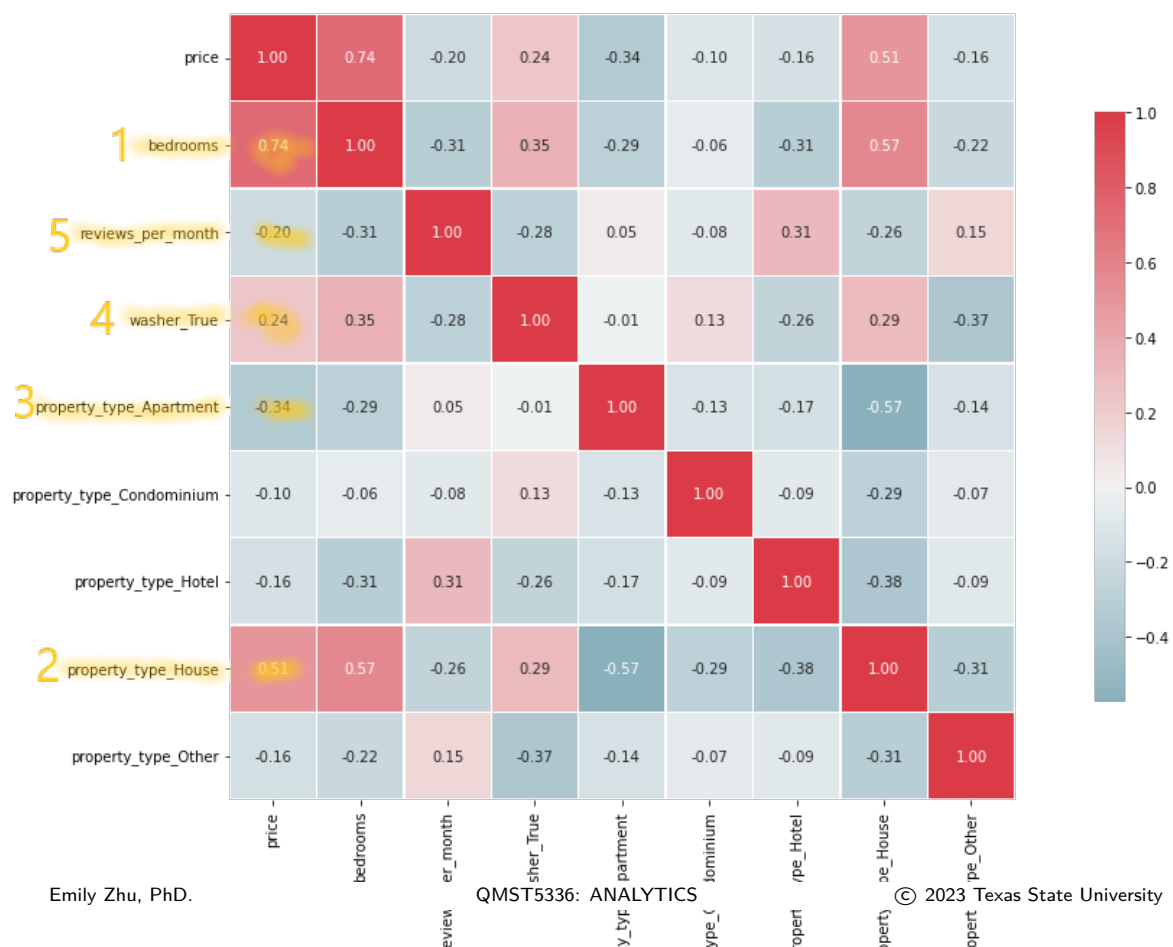
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Adjusted R Square: Application Example



Adjusted R Square: Application Example



Adjusted R square: Example C'd

Explanatory Variables	Adj. R^2	R^2
bedrooms	0.540	0.545
bedrooms + House	0.547	0.556
bedrooms + House + Apartment	0.550	0.564
bedrooms + House + Apartment + washer	0.545	0.564
bedrooms + House + Apartment + reviews	0.546	0.565

Conclusion: three explanatory variables are chosen:
bedrooms + House + Apartment.

Adjusted R square: Example C'd

Explanatory Variables	Adj. R^2	R^2
bedrooms	0.540	0.545
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Output Summary

OLS Regression Results

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	coef	std err	t	P> t	[0.025	0.975]
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Prob(Omnibus):	0.763	Jarque-Bera (JB):	0.110
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Warnings:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Population v.s. Sample: Regression

Think of:

- Line $y_i = b_0 + b_1x_i + e_i$ is an estimate of a regression line for the population.
- The population regression line is given by

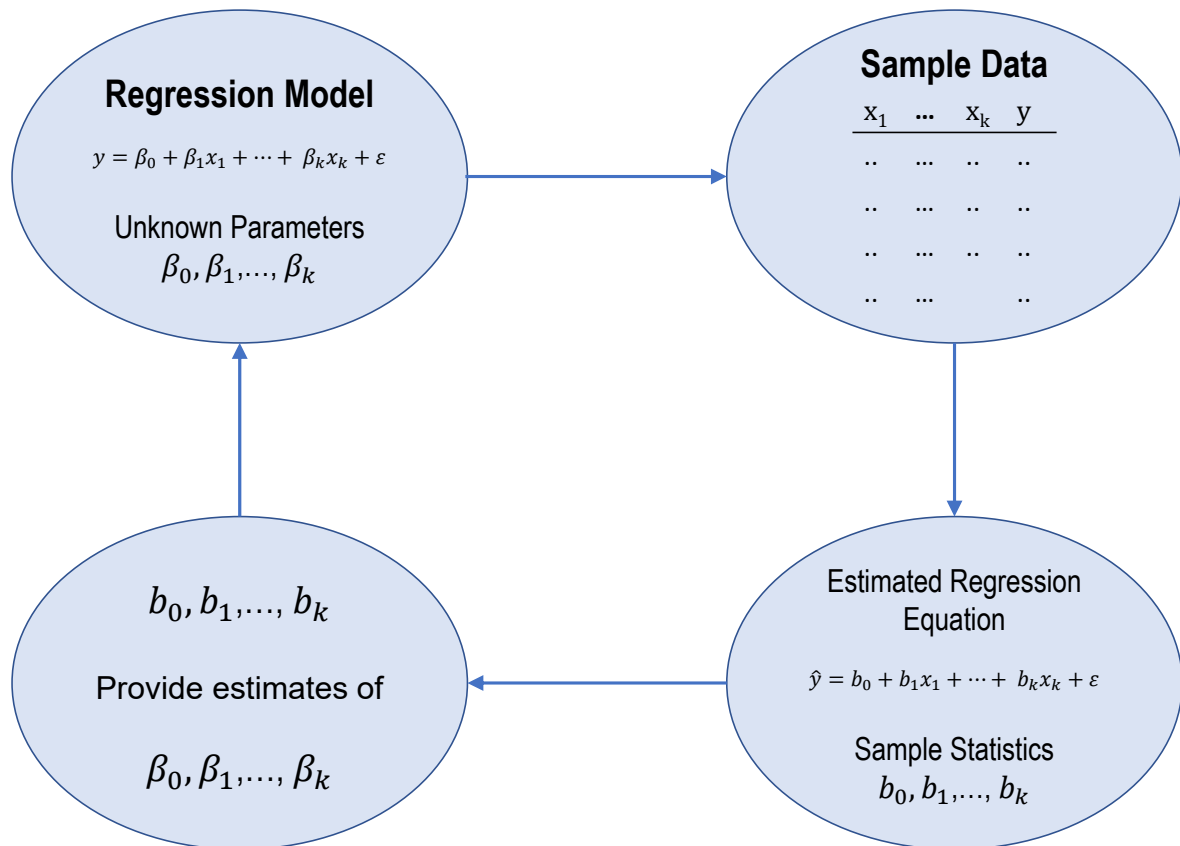
$$y_i = \beta_0 + \beta_1x_i + \epsilon_i$$

- β_0 and β_1 are parameters that describe the population
- ϵ_i are the residual errors
- b_0 and b_1 are statistics calculated from a sample
- b_0 is an estimate for β_0
- b_1 is an estimate for β_1

This is the same idea as

- $\bar{x}$ is the statistic that describes the population mean μ

Population v.s. Sample: Regression



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Regression: Model

Theorem

Let the regression line be

$$y_i = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_p x_{p,i} + \epsilon_i$$

The residuals ϵ_i are Normally distributed with mean 0 and standard deviation σ , $\epsilon \sim \mathcal{N}(0, \sigma)$. σ is known as the Standard Error.

But of course we don't know the exact model

- We don't know β_0
- We don't know β_1
- $\dots$
- We don't know β_p
- We don't know σ

We can estimate these value from our sample

- The estimate for β_0 is b_0 intercept
- The estimate for β_1 is b_1 coef of x_1
- $\dots$
- The estimate for β_p is b_p coef of x_p

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Analyzing Regression Estimates

To test how reliable our estimates of b_0 and $b_1, \dots, b_p$ are, we will use:

- Confidence Intervals [0.025 0.975]
- Hypothesis Tests t & $P > |t|$

Confidence Intervals: Coefficients

We only have a **single sample of data**, and not the **entire population of data**. So

- impossible to actually know the "true" coefficient is either within this interval or not
- We estimate the coefficient with the data we do have, and show uncertainty about that estimate by giving a range that the coefficient is probably within.

```
1 ### Statsmodels ###  
2 # print the confidence intervals for the model coefficients  
3 lm1.conf_int()
```

	0	1
Intercept	-28.508898	155.175564
Bedrooms	20.165041	83.168292

- The 95% [0.025 0.975] confidence intervals is just a convention
- You can create 90% confidence intervals (which will be more narrow)
- 99% confidence intervals (which will be wider)
- or whatever intervals you like.

Output Summary

OLS Regression Results

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Warnings:

Hypothesis Tests: Slope

In linear regression, we usually want to know whether **the relationship between y and x is significant or not**

- In other words, we will like to analyze the slope β_1
- Specifically our hypothesis looks like $\beta_1 = 0$

P-value

- the P in $P > |t|$ means the probability that $\beta_1 = 0$.
- the **smaller** P-value, the **more significant** the relationship between y and x .

t is a statistics used to calculate P-values.

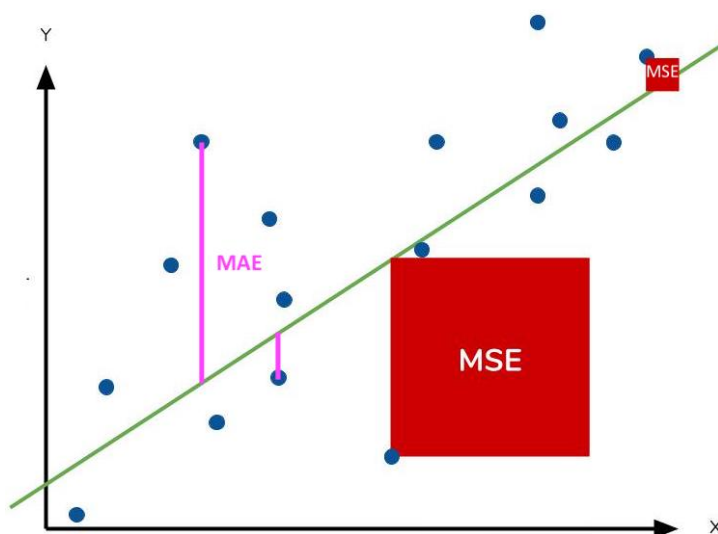
```
1 ### Statsmodels ###
2 # print the P-values for the model coefficients
3 lm1.pvalues
```

```
Intercept 0.142504
Bedrooms 0.007012
dtype: float64
```

Supervised Learning: Terminology

	Explanatory Variables					Dependent Variable
	Features 1	Features 2	Features 3	...	...	y
training set in-sample	152	25	63	...	...	85
	12	52	42	...	...	12
	...	...	...	...	...	...
	...	...	...	...	...	...
	...	...	...	...	...	...
	...	...	...	...	...	...
	...	...	...	...	...	...
testing set out-of-sample	...	...	...	...	...	...
	...	...	...	...	...	...
	...	...	...	...	...	...

How Good is the Prediction?



Measures of Prediction Errors

- MAE: mean absolute errors
- MSE: mean squared errors
 - better than MAE
 - penalize big errors
- RMSE: root mean squared errors
 - better than MSE
 - penalize big errors
 - the same unit as the dependent variable

Measures in Python

```
1 ### Scikit Learn###
2 from sklearn import metrics
3 # define true and predicted response values
4 y_true = [100, 50, 30, 20]
5 y_pred = [90, 50, 50, 30]
6
7 # calculate MAE, MSE, RMSE
8 print(metrics.mean_absolute_error(y_true, y_pred))
9 print(metrics.mean_squared_error(y_true, y_pred))
10 print(np.sqrt(metrics.mean_squared_error(y_true, y_pred)))
```

10.0
150.0
12.24744871391589

Evaluate Prediction Using Testing Set

```
1 ### Scikit Learn###
2 from sklearn.model_selection import train_test_split
3
4 # Prepare data
5 X = df_R[['bedrooms', 'accommodates', 'washer', 'beds', '
6         cancellation_policy_flexible', 'Review_Good']]
7 y = df_R.price
8
9 # Split data
10 X_train, X_test, y_train, y_test = train_test_split(X, y,
11                                                    random_state=1)
12
13 # Fit Model
14 lm2 = LinearRegression().fit(X_train, y_train)
15
16 # Predict
17 y_pred = lm2.predict(X_test)
18
19 # RMSE
20 print(np.sqrt(metrics.mean_squared_error(y_test, y_pred)))
```

Statistical Models V.S. Machine Learning

Statistical Models

- (adjusted) r^2
- hypothesis testig
- confidence interval
- sample & population
- proper model: why & how models work

Machine Learning

- training/testing sets
- MAE, MSE, RMSE
- emphasis on predicting
- models can be “black box”

However

They are NOT seperate. Regression models with low r^2 tend to have poor prediction power, i.e. big RMSE on testing sets.

Explanary Variables	R^2	RMSE
bedrooms	0.545	123.762
bedrooms + House	0.556	122.698
bedrooms + House + Apartment	0.564	121.924

Categorical Variables as Explanatory Variables

Example 1: Suppose we want to test the impact of property type on the price, where the categorical variable “property type” is

$$\text{property type} = \{\text{apartment, house}\}$$

Example 2: Suppose we want to test the impact of neighborhood on the price, where the categorical variable “Neighborhood” is

$$\text{Neighborhood} = \{\text{Hyde Park, Downtown, North loop}\}$$

We want to build a linear regression model of the type

$$\text{price} = b_0 + b_1(\text{Number of bedrooms}) + b_2(\text{property type}) + e$$

This model would make no sense. . .

Categorical Variables as Explanatory Variables

Idea: Introduce “Indicator variables” (sometimes referred to as dummy variables)

- Indicator variables is a variable that is either 0 or 1

Example 1: property type = {apartment, house}

We create the indicator variable $\text{House} = \{0 \text{ or } 1\}$ where

- $\text{House} = 1$ if the observation is a house
- $\text{House} = 0$ if the observation is an apartment

Categorical Variables as Explanatory Variables

Type of property	House (indicator)
house	1
house	1
house	1
house	1
house	1
house	1
house	1
apartment	0
house	1
apartment	0
apartment	0
apartment	0
house	1
apartment	0
house	1
house	1

The linear regression line becomes

$$\text{price} = b_0 + b_1(\text{Number of bedrooms}) + b_2(\text{House}) + e$$

How do we interpret b_2 ?

Categorical Variables as Explanatory Variables

The coefficient of a indicator variable is the effect on y , of having a characteristic, relative to not having it

$$\text{price} = b_0 + b_1(\text{Number of bedrooms}) + b_2(\text{House})$$

A house will be b_2 dollars more expensive than an apartment, all other things being equal

- Regression line for a house (house = 1)

$$\text{price} = b_0 + b_1(\text{Number of bedrooms}) + b_2(1)$$

- Regression line for apartment (house = 0)

$$\text{price} = b_0 + b_1(\text{Number of bedrooms}) + b_2(0)$$

Categorical Variables as Explanatory Variables

What if our categorical variable has more than 2 categories?

Example: Suppose we want to test the effect of reviews on listing prices, where

$$\text{Review} = \{\text{Good, Soso, Bad}\}$$

i.e., Review has 3 categories

If a category has M categories, we create $M - 1$ indicator variables

- Good = {1 or 0}
- Soso = {1 or 0}

Categorical Variables as Explanatory Variables

Example: How do reviews affect listing prices

$$\text{price} = b_0 + b_1(\text{Good}) + b_2(\text{Soso})$$

- 1 if observation is Good, 0 if not
- 1 if observation is Soso, 0 if not

What about Bad?

- If the observation's review is bad, then all indicators would be equal to 0
- We call “Bad” the excluded category

Review	Indicator for Good	Indicator for Soso
Good	1	0
Soso	0	1
Bad	0	0

Categorical Variables as Explanatory Variables

Question: Why do we have an excluded category?

Answer: Because we compare the effect of the categories with respect to the **excluded category**

Example: Suppose we run a regression and obtain the following result:

$$\text{price} = 100 + 100(\text{Good}) + 50(\text{Soso})$$

- prices are \$100 higher for good listings than bad ones
- prices are \$50 higher for soso listings than bad ones

Categorical Variables as Explanatory Variables

Example: Airbnb house listing prices in Austin

- **Dependent Variable:** Price (\$)
- **Independent Variables:**
Neighborhood = {Downtown, Hyde Park, North Loop}
 - excluded category: Downtown

```
1 ###Scikit-learn ###
2 X_cols = ['neighbourhood_hyde_park', 'neighbourhood_north_loop']

102.33333333333334
array([84.356666667 32.910000001])
```

- The average house in Hyde Park is 84.36\$ more expensive than in Downtown
- The average house in North Loop is 32.91\$ more expensive than in Downtown

Categorical Variables as Explanatory Variables

```
1 todummy_list = ['Review', 'instant_book']
2 #----function to generate dummy variables-----
3 def dummy_df(df, todummy_list):
4     for x in todummy_list:
5         dummies = pd.get_dummies(df[x], prefix=x, dummy_na=False)
6         df = df.drop(x, axis = 1)
7         df = pd.concat([df, dummies], axis=1)
8     return df
9 #----End of Fuction-----
10 df_R = dummy_df(df_sample, todummy_list)
11 df_R.head()
```

Review_Good	Review_Soso	Review_Bad	instant_book_t	instant_book_f
0	0	1	1	0
1	0	0	0	1
1	0	0	1	0
0	0	1	0	1
1	0	0	1	0

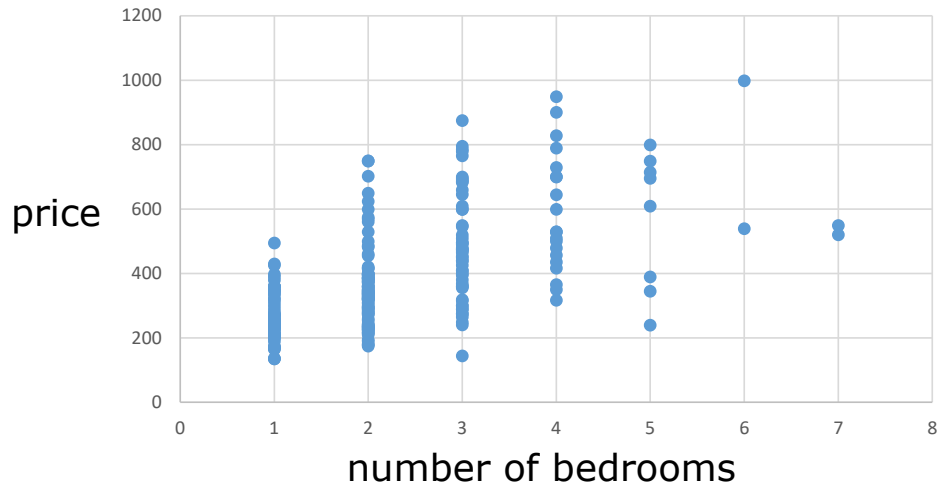
Note: It doesn't matter which category we exclude

- You can exclude whichever category you want
- The only thing that changes is the reference point
- Everything remains the same

Non-linear Relationship

Suppose a scatter plot reveals relationship is non-linear

Example: Relationship between number of bedrooms and price of the listing



Non-linear Relationship

To account for an exponential relationship

- we can create squared variable for independent variable x

Example:

$$\text{price} = b_0 + b_1(\text{Number of Bedrooms}) + b_2(\text{Number of Bedrooms})^2$$

Price	# bedrooms	# bedrooms squared
779	3	9
342	2	4
624	2	4
259.9	1	1
569	2	4
479	3	9
645	3	9
749	5	25
299	1	1
279	1	1
690	3	9
545	3	9
234	2	4
285	2	4
828	4	16
700	4	16
714	5	25
749.5	2	4
685	3	9

Treat Non-linear Relationship with Regression

Perform multiple regression in the usual way

Example: Regression output with and without the squared term

Visualize Regression in Handling Non-linear Relationship

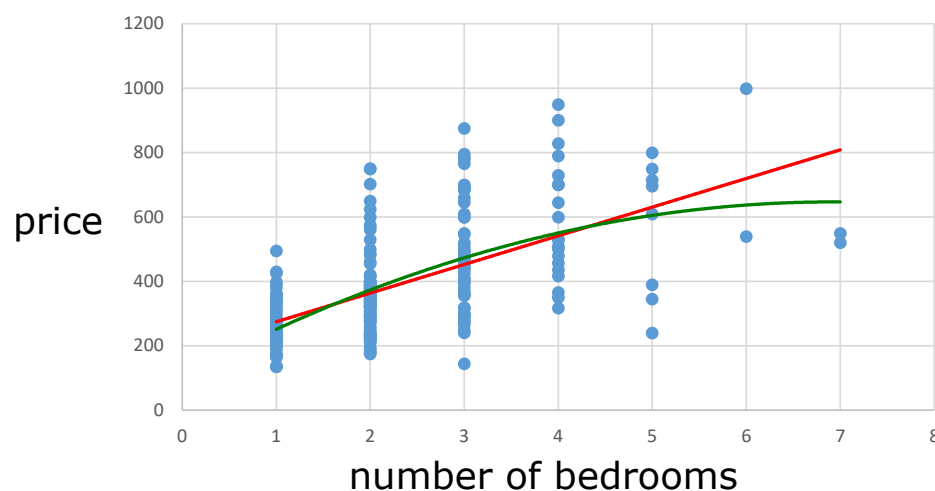
Example:

Linear

$$\text{price} = b_0 + b_1(\text{Number of Bedrooms})$$

Quadratic

$$\text{price} = b_0 + b_1(\text{Number of Bedrooms}) + b_2(\text{Number of Bedrooms})^2$$



Non-linear Relationship

This can be done for any non-linear relationship

- You can use logarithm $\ln(x)$
- trigonometric functions, e.g., $\sin(x)$, $\cos(x)$
- higher order polynomials, e.g., x^3 or x_1x_2

Procedure:

- First, look at the scatter plots between the dependent variable and all of predictors
- If the scatterplot shows non-linearities between two variables, we add a non-linear term
- If the non-linear term has a P-value higher than 0.1, we should not include it
- **Check RMSE on the testing set**



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Takeaway: ALWAYS Visualize your Data FIRST



https://en.wikipedia.org/wiki/Blind_men_and_an_elephant

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Before Next Lecture

- Assignment 2 has been posted on Canvas.

Clean & Visualize the data for your project

